

Project Scope Management

Topic # 4

Chapter 5 – Schwalbe

Chapter 5 – PMBOK

Scope

- **Product scope:** The features and functions that characterize a product, service, or result; and/or
- **Project scope:** The work performed to deliver a product, service, or result with the specified features and functions.
- The term project scope is sometimes viewed as including product scope.

Scope Statement

- Should be concise & clear.
- Vary with project types.
- The scope statement should include:
 - **A project Justification:** The business need that sparked the creation of the project
 - **Brief Description of Project's Products:** It summarizes the characteristics of products or services that the project will produce.
 - **Summary of Project Deliverables**
 - **Statement of Project Success Criteria:** A quantifiable criteria that must be met for project success.

Scope Statement

- **Project exclusion:** Explicitly stating what is out of scope for the project.
- **Constraints:** A limiting factor that affects the execution of a project or process. For example: a predefined budget or any imposed dates or schedule milestones, contractual provisions
 - Information on constraints may be listed in the project scope statement or in a separate log.
- **Assumptions:** A factor in the planning process that is considered to be true, real, or certain, without proof or demonstration.
 - Also describes the potential impact of those factors if they prove to be false.

Establishing project scope and objectives

- 1.1 Identify objectives and measures of effectiveness
 - 'how do we know if we have succeeded?'
- 1.2 Establish a project authority
 - 'who is the boss?'
- 1.3 Identify all stakeholders in the project and their interests
 - 'who will be affected/involved in the project?'

Setting objectives

- Answering the question '*What do we have to do to have a success?*'
- Need for a *project authority*
 - Sets the project scope
 - Allocates/approves costs
- Could be one person - or a group
 - Project Board
 - Project Management Board
 - Steering committee

Objectives

Informally, the objective of a project can be defined by completing the statement:

The project will be regarded as a success if.....

Rather like *post-conditions* for the project

Focus on *what* will be put in place, rather than *how* activities will be carried out

Objectives/Goal should be SMART

- S** – specific, that is, concrete and well-defined
- M** – measurable, that is, satisfaction of the objective can be objectively judged
- A** – achievable, that is, it is within the power of the individual or group concerned to meet the target
- R** – relevant, the objective must relevant to the true purpose of the project
- T** – time constrained: there is defined point in time by which the objective should be achieved

Goals/sub-objectives

These are steps along the way to achieving the objective. Informally, these can be defined by completing the sentence...

**Objective X will be achieved
IF the following goals are all achieved**
 A.....
 B.....
 C..... etc

Goals/sub-objectives continued

Often a goal can be allocated to an individual. Individual may have the capability of achieving goal, but not the objective on their own e.g.

Objective – user satisfaction with software product

Analyst goal – accurate requirements

Developer goal – software that is reliable

Measures of effectiveness

How do we know that the goal or objective has been achieved?

By a practical test, that can be objectively assessed.

e.g. for user satisfaction with software product:

- Repeat business – they buy further products from us
- Number of complaints – if low etc etc

Quantifying benefits

Benefits can be:

- Quantified and valued e.g. a reduction of x staff saving £ y
- Quantified but not valued e.g. a decrease in customer complaints by $x\%$
- Identified but not easily quantified – e.g. public approval for a organization in the locality where it is based

Scope Statement of a game

- **Project Title:** Bouncy Bunny Counts Money
- **Project Justification/ Business Need:** Our company has a hole in the market when it comes to children's items. *Bouncy Bunny Counts Money* will fulfill this hole and further boost sales of our accounting software market to parents of the children using this game.
- **Project Scope:** This project will consist of creating a marketable game based upon the *Bouncy Bunny Counts Money* comics on our website. The project will be completed by December, 2010. Modules of the game will include a simple shopping game, a way for Bouncy Bunny to earn money, and a way to motivate players to continue to play.

Scope Statement of a game

- **Project Deliverables:**
 - Scope Statement
 - Progress Reports
 - Issues Reports
 - Weekly Meeting Notes
 - Testing Strategy & Acceptance Testing Document
 - Final Game
 - User Guide
- **Project Success:** The project will be determined successful if the game sells.

Scope statement Example: Errors

- There might be way more deliverables than listed.
- Team Members & Important Stakeholders are not listed.
- It will be important for this project team to further define what project success will look like. How many copies should sell? How long should they take to sell?
- In the project justification, it is mentioned that this project is to also boost sales of company x's accounting software. How many units of that should be sold?

Scope statement Example: Errors

- In order to run a successful project, it will be important for stakeholders to see a timeline and to note important milestones. For instance, when will the game concept art be due? When will the scripting be complete? When will the product packaging be completed?
- There is no budget included. How much will it cost to create *Bouncy Bunny Counts Money*? How many employees will be required?

CASE STUDY

CASE ANALYSIS

SCOPE DISCUSSION OF A CAR POOLING APP

Requirement Gathering through Communications

- Interviews & Focus Groups
- Facilitated workshops
- User stories
- Questionnaires
- Observation
- Prototypes
- Benchmarks
- Existing System & Documents

Requirement Traceability Matrix (RTM)

- One method that can help track the requirements from the requirements document to the finished product is a *requirements traceability matrix (RTM)*.

Requirements Traceability Matrix							
Project Name:							
Cost Center:							
Project Description:							
ID	Associate ID	Requirements Description	Business Needs, Opportunities, Goals, Objectives	Project Objectives	WBS Deliverables	Product Design	Test Cases
001	1.0						
	1.1						
	1.2						
	1.2.1						
002	2.0						
	2.1						
	2.1.1						
	2.0						
003	3.1						
	3.2						
	4.0						
004	4.0						
005	5.0						

Work Breakdown Structure (WBS)

- A WBS is a decomposition of the work required to complete the project.
- A WBS is made up of work tasks at the lowest level.
- It helps to improve the accuracy of time, cost & resource estimates & helps to communicate clear work responsibilities.
- The WBS is a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables.
- The WBS organizes and defines the total scope of the project.

Creating WBS

- Decomposition is a technique used for dividing and subdividing the project scope and project deliverables into smaller, more manageable parts.
- The work package is the work defined at the lowest level of the WBS for which cost and duration can be estimated and managed.
- Expert judgment is often used to analyze the information needed to decompose the project deliverables down into smaller component parts in order to create an effective WBS.
- The project manager, in collaboration with the project team, then determines the final decomposition of the project scope into the discrete work packages that will be used to effectively manage the work of the project.

Why WBS?

- This helps you to objectively identify all of the work necessary without subconsciously leaving out real work in order to fit date constraints.
- The WBS step helps accomplish the following key objectives:
 - Develops an objective, rational view of the amount of work required
 - Helps team grasp the skills required and amount of resources required for the project
 - Provides a clear framework for assigning to individuals a clear task definition and delegate the responsibility for completion
 - Lays a foundation for analyzing the task dependencies and for isolating and managing risks
 - Lays a foundation for developing a bottom-up estimate for the project schedule
 - Allows tradeoffs to be made consciously and with the proper consensus involved.

Why WBS?

- A WBS defines the required way to complete the project:
- A WBS creates a sense of urgency
- A WBS can prevent scope creep
- A WBS provides control

WBS Structure & Types

- WBS can have two forms:
 - Tabular Form
 - Like an Organizational Chart (Diagrammatic Form)
- WBS can be made through
 - Functions or Process
 - Phases
 - Deliverables
 - Cost Structures
 - Business Functions

Projects and Activities

B - Identifying Activities

- **Work-based:** draw-up a Work Breakdown Structure listing the work items needed
- **Product-based approach**
 - list the deliverable and intermediate products of project – product breakdown structure (PBS)
 - Identify the order in which products have to be created
 - work out the activities needed to create the products
- **Hybrid approach:** A mix of the activity-based approach and the product-based approach.

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Approaches for Identifying Activities



Activity Based Approach

- The activity based approach consists of creating a list of all activities that the project is thought to involve.
- This can be done by
 - Brainstorming session involving the whole project team
 - The analysis of similar past projects.

Activity Based Approach

- When preparing the WBS:
 - Too great depth should be avoided as it will result in a large number of tasks that will be difficult to manage.
 - Too shallow structure should be avoided as this will provide insufficient detail for project control.
 - Each branch should be broken down at least to a level where each leaf may be assigned to an individual or responsible section within the organization.

Activity Based Approach

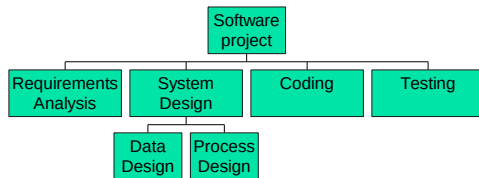
- Advantages of WBS are
 - More likely to obtain a task catalogue that is complete and composed of non-overlapping activities.
 - WBS represents a structure that can be refined as the project proceeds. It can start shallow early in the project and can be developed as information becomes available.

Product Based Approach

- It consists of producing a product breakdown structure PBS, and a product flow diagram PFD.
- Product Breakdown Structure (PBS)
 - It shows how a system can be broken down into different products.
 - It is less likely that a product will be left out of a PBS than that an activity might be omitted from an unstructured activity list. Why?

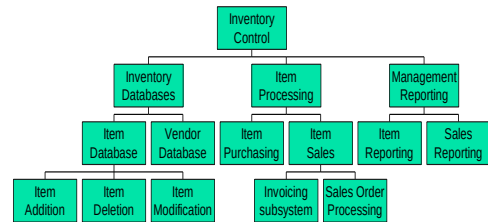
Work breakdown structure – An Example

Work Breakdown Structure (an extract)

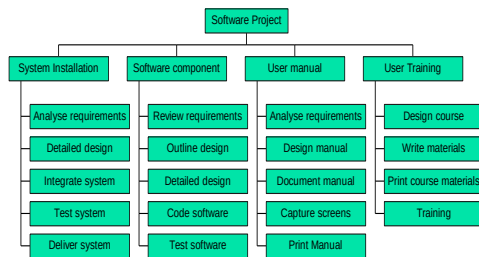


Product based approach - An example

A Product Breakdown Structure (an extract)



Hybrid approach – An example

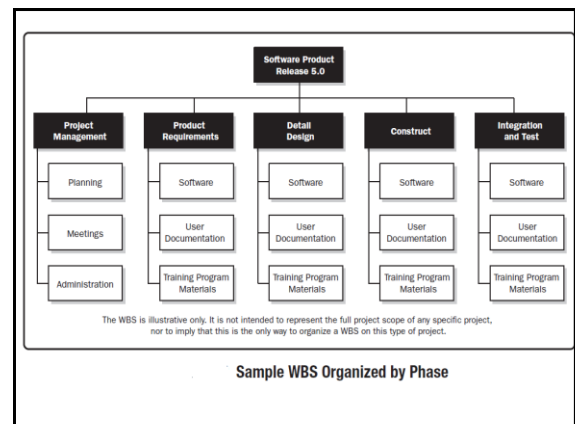
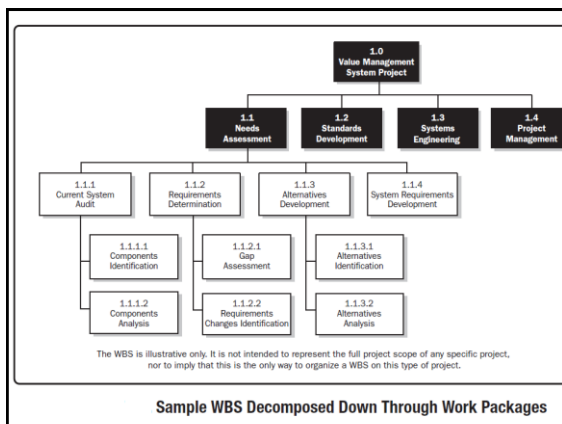


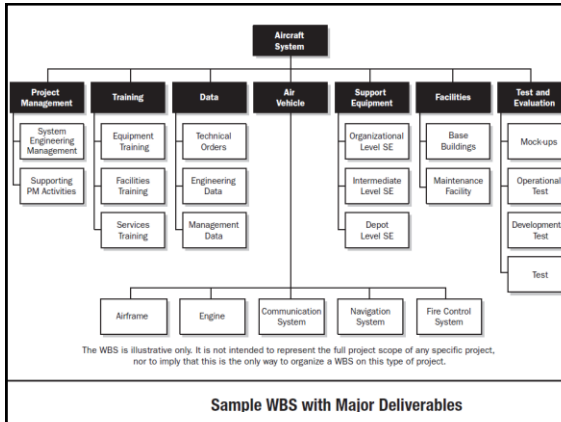
A Work Breakdown Structure based on deliverables

Hybrid Approach (cont'd)

- IBM in its MITP methodology suggests 5 levels
 - Level 1: Project
 - Level 2: Deliverables (software, manuals etc)
 - Level 3: Components
 - Level 4: Work-packages
 - Level 5: Tasks (individual responsibility)

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Example: WBS Organized by product

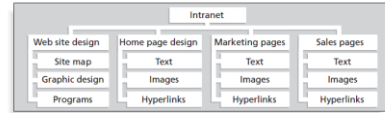
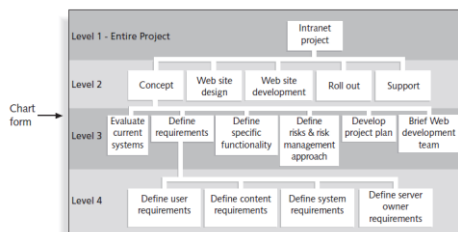


FIGURE 5-3 Sample intranet WBS organized by product

Example: WBS Organized by Phase



Example: WBS in tabular form

- 1.0 Concept
 - 1.1 Evaluate current systems
 - 1.2 Define requirements
 - 1.2.1 Define user requirements
 - 1.2.2 Define content requirements
 - 1.2.3 Define system requirements
 - 1.2.4 Define server owner requirements
 - 1.3 Define specific functionality
 - 1.4 Define risks and risk management approach
 - 1.5 Develop project plan
 - 1.6 Brief Web development team
- 2.0 Web site design
- 3.0 Web site development
- 4.0 Roll out
- 5.0 Support

END OF TOPIC 4A

- COMING UP!!!!!!
- Midterm Examination 1
- WBS continued
- Team Analysis
- Project Time Management